

PROJECT

Heimdall

System Architecture Documentation

An AI-driven SIEM orchestrator for automated Tier 1 SOC triage

TEAM MEMBERS

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1. Overview

Security teams receive far too many alerts. Most are low-priority noise, but a few are real threats — and finding those few by hand is slow and exhausting. This problem is called alert fatigue, and it causes real attacks to be missed.

Project Heimdall automates the first level of this work, known as Tier 1 triage. It connects a security-monitoring tool to an AI model that acts like a junior analyst: for each important alert, the AI explains what happened, rates how serious it is, and suggests what to do. The result is posted to the team's Slack channel as a short, color-coded card that a human can understand in a few seconds.

Heimdall does not replace the analyst. It handles the repetitive first pass so the human can focus on real decisions.

2. How It Works

Heimdall is a pipeline made of four stages. An alert moves through them in order:

Detection (Wazuh) — watches the computers and spots suspicious activity.

Orchestration (n8n) — receives the alert, removes the noise, and controls the flow.

Intelligence (the AI) — reads the alert and writes a clear verdict.

Notification (Slack) — shows the verdict to a human as a color-coded card.

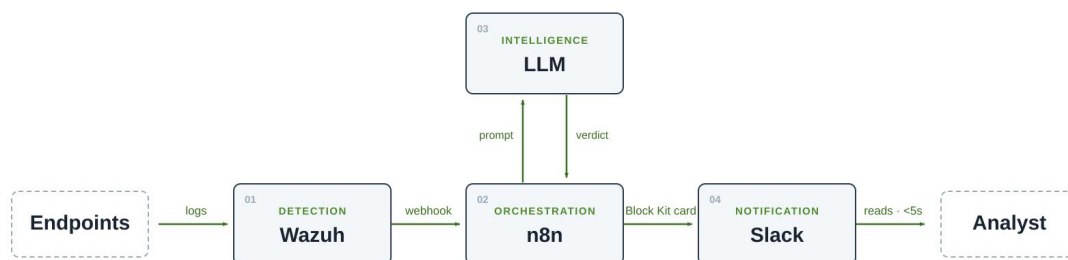


Figure 1 — The four-stage Heimdall pipeline.

n8n sits in the middle and acts as the coordinator: it sends the alert to the AI, receives the answer, and then posts the final card to Slack.

3. The Four Components

Wazuh — Detection. An open-source security-monitoring tool (a SIEM). It runs on the computers we watch, notices suspicious behavior — for example, many failed login attempts — and sends out an alert.

n8n — Orchestration. A self-hosted automation tool. It catches the alert, drops low-priority noise, prepares the data, calls the AI, and sends the final result to Slack.

The AI (LLM) — Intelligence. A large language model acts as a digital analyst. It reads the raw alert and returns a short, structured verdict: a summary, the matching MITRE ATT&CK category, a severity level, and recommended actions. The specific model can be swapped (for example, OpenAI or DeepSeek).

Slack — Notification. The team's chat app. Heimdall posts a color-coded card — red, amber, or green by severity — so an analyst can react quickly.

4. How an Alert Flows Through the System

Here is what happens for a single alert — for example, someone trying many passwords over SSH:

1. Wazuh notices the repeated failed logins and creates an alert.
2. n8n receives the alert and drops it if it is only low-level noise.
3. n8n sends the important alert to the AI.
4. The AI summarizes the threat, tags it (MITRE ATT&CK T1110 – Brute Force), rates it HIGH, and suggests actions.
5. n8n turns this into a color-coded Slack card.
6. An analyst reads the card and decides what to do — in under five seconds.

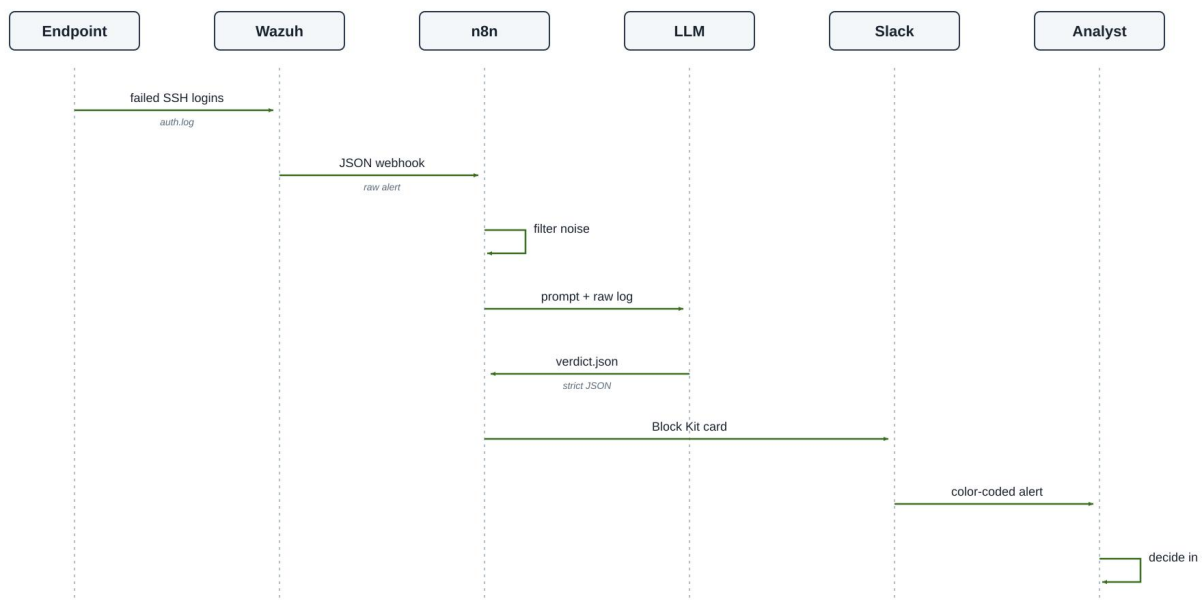


Figure 2 — An alert traced from the endpoint to the analyst.

5. Technology Stack

Heimdall is built mostly from open-source, self-hosted parts. The only external service is the AI model.

Tool	Role
Wazuh	Detection — open-source SIEM and endpoint monitoring.
n8n	Orchestration — self-hosted workflow automation.
LLM	Intelligence — AI analyst (OpenAI / DeepSeek, swappable).
Slack	Notification — color-coded alert cards for the team.

6. Security and Privacy

Because Heimdall handles security data, a few points are important:

- Alert data is sent to an external AI service, so include only what is needed and check the provider's data policy.
- Keep API keys and tokens in n8n's secure settings — never in shared files.
- Protect the n8n webhook so fake alerts cannot be injected.
- The human review step is also a safeguard: the AI can be wrong, so a person always makes the final call.

7. Limitations and Future Work

Current limits. Heimdall runs on a small set of test computers, not a full network. It suggests actions but does not carry them out automatically, and the quality of each verdict depends on the AI model.

Future ideas. Automatic response for clear, low-risk cases; a local AI model so data stays in-house; and a dashboard showing how many alerts were handled and how quickly.